Common Problems

What are cryptocurrencies?

Bitcoin, Ethereum, and other cryptocurrencies are revolutionizing the way we invest, bank, and use money. Cryptocurrency is essentially a **decentralized digital currency** designed specifically for use on the internet. Bitcoin was the first cryptocurrency, launched in 2008, and remains the largest, most influential, and best-known cryptocurrency. In the decade since, Bitcoin and other cryptocurrencies such as Ethereum have evolved into digital alternatives to government-issued currencies.

The most popular cryptocurrencies by market capitalization are Bitcoin, Ethereum, Tether, and Solana. Other well-known cryptocurrencies include Tezos, EOS, and ZCash. Some cryptocurrencies are similar to Bitcoin. Others are based on different technology or have new functions beyond just transferring value.

> **Efficiency:** Cryptocurrencies allow value to be transferred online without the need for middlemen like banks or payment processors, allowing value to be transferred almost instantly around the clock around the world with low fees.

Cryptocurrencies are not usually issued or controlled by any government or other central authority. They are managed by a peer-to-peer network of computers running free, open-source software. Generally, anyone who wants to can participate.

Security and the Blockchain

If there are no banks or governments involved, how can cryptocurrency be secure? Cryptocurrency is secure because all transactions are vetted through a technology called **blockchain**.

A cryptocurrency blockchain is similar to a bank's balance sheet or ledger. Each currency has its own blockchain, which is a continually revalidated record of every transaction made using that currency. Unlike a bank ledger, a cryptocurrency blockchain is distributed among participants across the entire network of digital currencies. No company, country, or third party has control; anyone can participate.

Key Concepts

Transferability

Cryptocurrencies make transactions between people who are far apart as seamless and convenient as paying with cash at your local grocery store. Sending bitcoin is typically 60x more cost-effective and up to 48x faster than an international wire transfer.

Privacy

When paying with cryptocurrency, you don't need to provide merchants with unnecessary personal information. This means your financial information is protected and not shared with third parties such as banks, payment services, advertisers, and credit rating agencies. Because sensitive information doesn't need to be sent over the internet, there's little risk of your financial information being compromised or your identity being stolen.

Security

Almost all cryptocurrencies, including Bitcoin, Ethereum, Tezos, and Bitcoin Cash, are secured using a technology called blockchain, which is constantly checked and verified by massive amounts of computing power. The network that supports Bitcoin has never been hacked. The basic concept of cryptocurrency makes it secure: the system is permissionless and its core software is open source, allowing countless computer scientists and cryptographers to examine every aspect of the network and its security.

Portability

Because your crypto assets are not tied to a financial institution or government, you can use your crypto no matter where you are or what happens to the major intermediaries of the global financial system.

Transparency

Every transaction on the Bitcoin, Ethereum, Tezos, and Bitcoin Cash networks is published publicly without exception. This means there is no room for manipulating transactions, changing the currency supply, or adjusting the rules mid-run.

Irreversibility

Unlike credit card payments, cryptocurrency payments cannot be reversed. For merchants, this significantly reduces the likelihood of fraud. For customers, it has the potential to eliminate important disputes that credit card companies charge high processing fees for, making commercial transactions cheaper.

Why Cryptocurrency is the Future of Finance

Cryptocurrencies are the first alternative to the traditional banking system, offering powerful advantages over previous payment methods and traditional asset classes. Think of them as Money 2.0 - a new type of cash unique to the Internet, making it possible to become the fastest, easiest, cheapest, most secure, and most ubiquitous way to exchange value in the world's history.

Global Economic Freedom

Digital currencies provide equal opportunity, no matter where you were born or live. As long as you have a smartphone or other internet-connected device, you have the same access to cryptocurrency as everyone else. They create a unique opportunity to expand people's global economic freedom. The fundamentally borderless nature of digital currencies promotes free trade even in countries where governments exercise strict control over citizens' finances.

What are stablecoins?

USD Coin is an example of a cryptocurrency called a **stablecoin**. You can think of it as a crypto dollar. USD Coin's purpose is to minimize volatility and maximize utility. Stablecoins offer some of the best attributes of cryptocurrencies (seamless global transactions, security, and privacy) with the valuation stability of fiat currencies.

Stablecoins work by pegging their value to an external factor, typically a fiat currency like the U.S. dollar or a commodity like gold. This stability can increase their utility as money for everyday use, as buyers and merchants can be confident that the value of their transactions will remain relatively consistent.

How do cryptocurrencies work?

Bitcoin is the first and most famous cryptocurrency, but there are now thousands of them. Many, like Litecoin and Bitcoin Cash, share Bitcoin's core features but explore new ways to process transactions. Others offer broader functionality. Ethereum, for example, can be used to run applications and create contracts. But all four are based on a concept called the **blockchain**.

Blockchain technology is also exciting because it has many uses beyond cryptocurrency. Blockchain is being used to explore medical research, improve medical record sharing, streamline supply chains, increase privacy on the internet, and more.

Public-Private Key Cryptography

Cryptocurrencies use a technique called public-private key cryptography to transfer ownership of money to a secure distributed ledger. A **private key** is a super-secure password that never needs to be shared with anyone. The associated **public key** can be freely and securely shared with others to receive value on the network. It is impossible for anyone to guess your private key using your public key.

Cryptocurrency Mining

Most cryptocurrencies are "mined" through a decentralized network of computers. But mining doesn't just generate more Bitcoin or Ethereum; it's also a mechanism that updates and secures the network by continually validating the public blockchain ledger and adding new transactions.

Technically, **anyone with a computer and an internet connection can become a miner**. However, mining isn't always profitable. Depending on the cryptocurrency you're mining, the speed of your computer, and the cost of electricity in your area, you could end up spending more on mining than you earn from the cryptocurrency. As a result, most cryptocurrency mining is currently done by specialized companies or large groups of people (mining pools) who contribute their combined computing power.

How does the network encourage miners to participate? Taking Bitcoin as an example, the network uses a "lottery" system where mining devices around the world compete to be the first to solve a complex mathematical problem. The winner verifies and updates the blockchain with new transactions and receives newly created Bitcoins as a reward.

Important Questions

Where does the value of cryptocurrency come from?

Like all goods and services, the economic value of cryptocurrencies comes from **supply and demand**. Supply refers to the amount available (e.g., the limited number of Bitcoins that will ever exist). Demand refers to people's desire to own it and how eager they are to do so. The value will always be a balance of these two factors.

There are also other types of value. Many people find value in supporting an exciting new financial system. Similarly, some enjoy using Bitcoin because of its low fees or to encourage businesses to adopt it.

How to Buy and Store Cryptocurrency

The easiest way to get cryptocurrency is to buy it on an online exchange like Coinbase. You can buy major coins like Bitcoin (BTC), Litecoin (LTC), Ethereum (ETH), and Bitcoin Cash (BCH), or explore emerging coins like Stellar Lumens or EOS. You don't need to buy a whole coin; most exchanges allow you to buy fractions of a coin for as little as $2.

Storing cryptocurrency is similar to storing cash—you need to protect it from theft and loss. While there are many ways to store it both online and offline, using a trusted exchange is often the simplest solution for beginners. Customers can securely store, send, receive, and trade cryptocurrency by logging into their account on a computer, tablet, or phone.

What can you do with cryptocurrency?

Cryptocurrency features are expanding rapidly. Here are a few ways to get started:

- **Shopping:** Over 8,000 merchants worldwide accept crypto through platforms like Coinbase Commerce.
- **Global Travel:** Using cryptocurrencies for travel can reduce currency exchange fees. A community of "crypto nomads" exists who use digital assets exclusively while traveling.
- **Charitable Donations:** Many non-profits now accept Bitcoin donations, offering a transparent way to give.
- **Gift Giving:** Cryptocurrencies make unique gifts for those interested in learning about new technologies.
- **Tipping:** Creators and musicians often leave a Bitcoin address or QR code to allow fans to show appreciation.
- **Discover New Tech:** Explore combinations of currency and technology, like Orchid (a VPN that uses the OXT cryptocurrency).
- **Virtual Worlds:** Buy property or items in virtual gaming worlds like Decentraland, which runs on the Ethereum blockchain.

- **DeFi (Decentralized Finance):** Explore mutual fund-like investments and loan lending mechanisms without the need for a central authority.